

Master Descriptive Statistics Reference

India Time Use Survey 2024 - Comprehensive Activity and Demographic Patterns

Research Team

November 13, 2025

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1 Executive Summary

This document provides comprehensive descriptive statistics for India's Time Use Survey 2024 to inform research question selection. It includes:

- **Participation rates** for all major activity codes
- **Time spent** conditional on participation (mean, median, range)
- **Sample sizes** for key demographic subgroups
- **Household composition** patterns
- **Cross-tabulations** by gender, marital status, age, education, and urban/rural status

Key Use: Consult this document BEFORE designing a research question to assess: 1. Whether sufficient people participate in the activity of interest 2. What demographic variation exists 3. What sample sizes are available for subgroup analyses

2 Data Overview

```
## Loading data and checking column names...
```

```
## Total columns available: 49
```

```
## WARNING: No household size column found.
```

```
##          used (Mb) gc trigger    (Mb) limit (Mb)  max used   (Mb)
## Ncells  811145 43.4    2242992  119.8          NA    2242992  119.8
## Vcells 1654956 12.7   653913768 4989.0        16384 503331072 3840.2
```

```
##          used (Mb) gc trigger    (Mb) limit (Mb)  max used   (Mb)
## Ncells   893375 47.8    2242992  119.8          NA    2242992  119.8
## Vcells 18100475 138.1 418504812 3193.0        16384 503331072 3840.2
```

Table 1: Dataset Overview (10% Random Sample with Adjusted Weights)

Metric	Value
Total Observations (10% sample)	835,346
Unique Households	137,005
Unique Individuals	394,405
Average observations per person	2.1
Survey Period	January-December 2024

3 Activity Code Participation Rates

3.1 All Major Activity Categories

Table 2: Participation Rates for All Activity Codes (Sorted by Participation Rate)

Code	Activity	Participation (%)	N People	Mean (min)	Median (min)
92	Eating/drinking	17.29	144766	32.71	30
91	Personal hygiene	13.27	110342	222.08	180
93	Sleeping/napping	10.74	89778	34.25	30
31	Food preparation	9.42	77930	49.54	30
71	Physical exercise/sports	7.51	61154	54.73	30
84	Attending ceremonies	7.13	60849	60.50	60
85	Civic participation	4.48	38309	54.94	30
18	NA	3.57	29381	35.39	30
41	Physical childcare	3.09	25913	48.12	30
12	NA	3.02	23689	153.16	150
32	Household upkeep	2.60	21636	40.88	30
61	Participating in culture	2.02	16765	134.83	150
74	Sports events attendance	1.89	16850	40.27	30
34	Shopping/procurement	1.62	13370	50.38	60
62	Attending cultural events	1.59	13276	90.30	90
13	NA	1.57	13674	163.50	150
82	Socializing remotely	1.55	12621	86.39	90
64	Hobbies/crafts	1.54	12925	33.86	30
21	Formal education/schooling	0.93	7537	77.73	60
11	NA	0.91	7744	165.66	180
83	Participating in events	0.64	5451	54.69	60
37	NA	0.50	4183	63.76	60
24	Library/learning resources	0.48	3801	47.14	30
75	Sports travel	0.38	3473	50.55	30
38	NA	0.36	2972	35.30	30
94	Healthcare	0.19	1557	46.32	30
14	NA	0.17	1366	36.66	30
25	Learning new skills	0.17	1393	32.60	30
63	Visiting heritage sites	0.15	1289	120.35	120
95	Receiving personal services	0.13	972	38.71	30
39	NA	0.12	1017	66.44	60
35	Household management	0.11	826	64.95	60
72	Outdoor recreation	0.10	819	135.06	120
44	Transporting children	0.10	906	43.45	30
36	NA	0.08	730	49.24	30
99	Other personal care	0.08	642	54.44	30
86	Organizational meetings	0.08	653	41.84	30
33	Care for textiles/clothing	0.07	601	62.83	60
51	Physical care adults	0.06	483	69.30	60
42	Teaching/supervising children	0.05	437	45.14	30
22	Homework/assignments	0.05	544	84.19	90
81	Socializing in person	0.04	366	104.42	90
89	Other social activities	0.04	338	82.06	60
16	NA	0.02	153	114.88	120
69	Other culture/leisure	0.02	158	104.93	90

23	Additional classes	0.02	134	143.97	150
43	Accompanying children	0.02	136	56.70	30
54	Accompanying adults	0.01	118	57.66	30
49	Other childcare	0.01	105	61.38	60
79	Other sports/outdoor	0.01	129	107.79	120
52	Healthcare support adults	0.01	95	113.23	120
73	Hunting/fishing	0.00	33	68.44	60
53	Administrative help adults	0.00	20	173.06	150
59	Other adult care	0.00	24	119.70	105
15	NA	0.00	19	138.09	150
17	NA	0.00	12	78.01	45

3.2 Key Activity Codes for Research Questions

3.2.1 Educational Activities (Codes 21-29)

Table 3: Educational Activities Detail (Codes 21-29)

Code	Activity	Part. (%)	N	Mean	Median	P25	P75
21	Formal education/schooling	0.93	7537	77.73	60	30	90
22	Homework/assignments	0.05	544	84.19	90	60	120
23	Additional classes	0.02	134	143.97	150	90	210
24	Library/learning resources	0.48	3801	47.14	30	30	60
25	Learning new skills	0.17	1393	32.60	30	30	30

3.2.2 Physical Exercise and Sports (Code 71)

Table 4: Physical Exercise and Sports (Code 71) - Summary Statistics

	Value
N_obs	61259.00
N_people	61154.00
Participation_pct	7.51
Mean_min	54.73
Median_min	30.00
SD_min	34.98
Min_min	30.00
Max_min	390.00

3.2.3 Religious and Spiritual Activities (Code 9)

Table 5: Religious and Spiritual Activities (Code 9) - Summary Statistics

	Value
N_obs	0
N_people	0
Participation_pct	0
Mean_min	NaN
Median_min	NA
SD_min	NA
Min_min	Inf
Max_min	-Inf

4 Demographic Breakdowns

4.1 Overall Demographics

Table 6: Gender Distribution

gender_label	N	Percent
Female	200729	53.44
Male	193633	46.55
Transgender	43	0.01

Table 7: Marital Status Distribution

marital_label	N	Percent
Currently married	239298	62.50
Divorced/Separated	2002	0.47
Never married	125792	30.04
Widowed	27313	6.99

Table 8: Age Group Distribution

age_group	N	Percent
18-24	49816	12.84
25-34	84750	22.08
35-44	66226	16.80
45-54	50837	13.01
55-64	40151	10.32
65+	30419	7.56
<18	72206	17.39

Table 9: Education Level Distribution

education_level	N	Percent
Higher education	55747	13.12
Primary or less	170266	44.72
Secondary	168392	42.16

Table 10: Urban/Rural Distribution

Sector	N	Percent
Rural	247802	68.85

Urban	146603	31.15
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4.2 Activity Participation by Gender and Marital Status

4.2.1 Educational Activities (21-29) by Demographics

Table 11: Educational Activities by Gender and Marital Status

Gender	Marital Status	N People	Part. (%)	Mean (min)
Female	Currently married	6086	4.96	62.41
Female	Divorced/Separated	43	3.76	64.88
Female	Never married	716	1.37	61.21
Female	Widowed	854	4.19	63.27
Male	Currently married	4173	3.84	75.35
Male	Divorced/Separated	21	2.72	42.39
Male	Never married	769	1.04	73.87
Male	Widowed	250	4.15	75.51
Transgender	Currently married	1	4.91	30.00
Transgender	Never married	0	0.00	NaN
Transgender	Widowed	0	0.00	NaN

4.2.2 Physical Exercise (71) by Demographics

Table 12: Physical Exercise by Gender and Marital Status

Gender	Marital Status	N People	Part. (%)	Mean (min)
Female	Currently married	19731	16.06	54.36
Female	Divorced/Separated	165	15.06	57.55
Female	Never married	6184	11.97	51.58
Female	Widowed	4367	21.25	60.67
Male	Currently married	20123	18.23	54.72
Male	Divorced/Separated	151	18.58	61.54
Male	Never married	9180	12.70	53.91
Male	Widowed	1247	21.78	65.37
Transgender	Currently married	4	20.01	53.46
Transgender	Never married	2	15.95	164.14
Transgender	Widowed	0	0.00	NaN

4.2.3 Religious Activities (9) by Demographics

Table 13: Religious Activities by Gender and Marital Status

Gender	Marital Status	N People	Part. (%)	Mean (min)
Female	Currently married	0	0	NaN
Female	Divorced/Separated	0	0	NaN

Female	Never married	0	0	NaN
Female	Widowed	0	0	NaN
Male	Currently married	0	0	NaN
Male	Divorced/Separated	0	0	NaN
Male	Never married	0	0	NaN
Male	Widowed	0	0	NaN
Transgender	Currently married	0	0	NaN
Transgender	Never married	0	0	NaN
Transgender	Widowed	0	0	NaN

5 Household Composition Patterns

5.1 Relationship to Household Head

Table 14: Distribution of Household Members by Relationship to Head

Relationship to Head	N People	Percent (%)
Self (Household Head)	120729	30.31
Spouse of Head	93230	26.04
Married Child	22886	5.31
Spouse of Married Child	24484	6.60
Unmarried Child	100639	24.11
Grandchild	14171	3.23
Father/Mother/Father-in-law/Mother-in-law	9915	2.58
Other Relative	7880	1.75
Servant/Employee/Other Non-Relative	471	0.08

5.2 Household Head Characteristics

Table 15: Age Distribution of Household Heads

	Value
Mean_age	46.7
Median_age	46.0
Min_age	6.0
Max_age	118.0

5.3 Children in Households

Table 16: Children (Age 6-18) by Gender of Household Head

Gender of Household Head	N Children (6-18)	Percent (%)
Female	9984	12.89
Male	56092	73.38
Transgender	2	0.00
NA	10654	13.73

6 Cross-Tabulations for Research Questions

6.1 Q1: Children’s Educational Activities by Household Head Gender

Table 17: Children’s Educational Activity Participation Rate by HH Head Gender

Gender of HH Head	N Children	Participation (%)
Female	9984	0.88
Male	56092	0.88
Transgender	2	0.00
NA	10654	0.87

6.2 Q2: Exercise by Gender and Marital Status (Marriage Penalty Analysis)

Table 18: Exercise Participation: Marriage Penalty Analysis

Gender	Marital Status	N People	Part. (%)	Mean (min)
Female	Currently married	125703	16.06	8.73
Female	Never married	52597	11.97	6.17
Male	Currently married	113570	18.23	9.98
Male	Never married	73181	12.70	6.84

6.3 Q3: Religious Activities by Gender and Marital Status

Table 19: Religious Activities: Gender and Marriage Patterns

Gender	Marital Status	N People	Part. (%)	Mean (min)
Female	Currently married	125703	0	0
Female	Never married	52597	0	0
Male	Currently married	113570	0	0
Male	Never married	73181	0	0

7 Additional Activity Breakdowns

7.1 Leisure and Recreation Activities (61-69)

Table 20: Leisure and Recreation Activities (Codes 61-69)

Code	Activity	Part. (%)	N	Mean	Median
61	Participating in culture	2.02	16776	134.83	150
62	Attending cultural events	1.59	13276	90.30	90
63	Visiting heritage sites	0.15	1289	120.35	120
64	Hobbies/crafts	1.54	12925	33.86	30
69	Other culture/leisure	0.02	158	104.93	90

7.2 Social Activities (81-89)

Table 21: Social Life and Communication Activities (Codes 81-89)

Code	Activity	Part. (%)	N	Mean	Median
81	Socializing in person	0.04	366	104.42	90
82	Socializing remotely	1.55	12621	86.39	90
83	Participating in events	0.64	5451	54.69	60
84	Attending ceremonies	7.13	60882	60.50	60
85	Civic participation	4.48	38310	54.94	30
86	Organizational meetings	0.08	653	41.84	30
89	Other social activities	0.04	338	82.06	60

7.3 Volunteer and Community Service (51-59)

Table 22: Volunteer and Community Service Activities (Codes 51-59)

Code	Activity	Part. (%)	N	Mean	Median
51	Physical care adults	0.06	484	69.30	60
52	Healthcare support adults	0.01	95	113.23	120
53	Administrative help adults	0.00	20	173.06	150
54	Accompanying adults	0.01	118	57.66	30
59	Other adult care	0.00	24	119.70	105

8 Age-Specific Patterns

8.1 Educational Activities by Age Group

Table 23: Educational Activities by Age Group

Age Group	N People	Part. (%)	Mean (min)
18-24	1181	2.46	62.62
25-34	2617	3.18	64.05
35-44	2867	4.51	66.84
45-54	2489	5.22	68.02
55-64	2083	5.27	73.19
65+	1140	3.75	71.52
<18	536	0.76	66.91

8.2 Exercise by Age Group and Gender

Table 24: Exercise by Age Group and Gender

Age Group	Gender	N People	Part. (%)	Mean (min)
18-24	Female	4038	16.09	54.60
18-24	Male	4024	16.97	57.30
25-34	Female	5981	13.83	52.21
25-34	Male	6148	15.66	52.29
35-44	Female	5077	15.48	52.80
35-44	Male	5239	16.72	53.16
45-54	Female	4564	17.89	54.88
45-54	Male	4349	18.42	53.24
55-64	Female	3938	19.20	58.16
55-64	Male	3853	20.59	57.65
65+	Female	3586	24.16	63.17
65+	Male	3659	24.69	64.47
<18	Female	3263	9.71	49.10
<18	Male	3429	9.19	48.78

8.3 Religious Activities by Age Group and Gender

Table 25: Religious Activities by Age Group and Gender

Age Group	Gender	N People	Part. (%)	Mean (min)
18-24	Female	0	0	NaN
18-24	Male	0	0	NaN
25-34	Female	0	0	NaN
25-34	Male	0	0	NaN

35-44	Female	0	0	NaN
35-44	Male	0	0	NaN
45-54	Female	0	0	NaN
45-54	Male	0	0	NaN
55-64	Female	0	0	NaN
55-64	Male	0	0	NaN
65+	Female	0	0	NaN
65+	Male	0	0	NaN
<18	Female	0	0	NaN
<18	Male	0	0	NaN

9 Urban-Rural Patterns

9.1 Key Activities by Urban/Rural Status

Table 26: Key Activities by Urban/Rural Status

Activity	Codes	Rural Part. (%)	Urban Part. (%)	Rural Mean	Urban Mean
Educational activities	21-29	4.49	0.87	68.66	54.37
Physical exercise	71	16.72	14.01	56.03	51.70
Religious activities	9	0.00	0.00	NaN	NaN

10 Recommendations for Research Question Selection

10.1 Summary of Feasibility

Based on the descriptive statistics, here are recommendations for the three proposed research questions:

10.1.1 1. Children's Educational Activities & Household Head Gender

Feasibility: Consult the tables in Section 6.1 for: - Sample sizes of children in female-headed vs male-headed households - Participation rates in educational activities - Mean time spent when participating

Decision criteria: - If participation rate > 10%: HIGH FEASIBILITY - If participation rate 5-10%: MODERATE FEASIBILITY (proceed with caution) - If participation rate < 5%: LOW FEASIBILITY (consider alternative specification)

10.1.2 2. Exercise and Marriage Penalty

Feasibility: Consult the tables in Section 6.2 for: - Exercise participation by gender and marital status - Sample sizes for never-married vs married individuals - Gender gaps in participation

Decision criteria: - If participation rate > 2%: FEASIBLE for marriage penalty analysis - If participation rate 1-2%: MARGINAL (consider combining with broader recreation codes) - If participation rate < 1%: NOT FEASIBLE (too few observations)

10.1.3 3. Religious Activities and Gender

Feasibility: Consult the tables in Section 6.3 for: - Religious activity participation by gender and marital status - Sample sizes for key comparisons - Age and education patterns

Decision criteria: - If participation rate > 5%: HIGH FEASIBILITY - If clear gender differences exist: STRONG research question - If age/education/marital variation exists: Multiple testable hypotheses

10.2 General Guidelines

Minimum participation rates for robust analysis: - Main effects (gender, marital status): 2-3% participation minimum - Interaction effects (gender $\times$ marital $\times$ age): 5-7% participation recommended - Triple interactions: 10%+ participation ideal

Sample size considerations: - For regression analysis with state fixed effects: 10,000+ observations ideal - For subgroup analysis: 1,000+ observations per subgroup - For rare activities: May need to pool age groups or simplify specifications

Statistical power: - Larger participation rates = more precise estimates - Lower participation rates = need larger effect sizes to detect

11 Appendix: Full Activity Code List

Table 27: Complete Activity Code Reference (All Codes)

Code	Activity	Part. (%)	N Obs	N People	Mean	Median
11	NA	0.91	7746	7744	165.66	180
12	NA	3.02	23693	23689	153.16	150
13	NA	1.57	13684	13674	163.50	150
14	NA	0.17	1366	1366	36.66	30
15	NA	0.00	19	19	138.09	150
16	NA	0.02	153	153	114.88	120
17	NA	0.00	12	12	78.01	45
18	NA	3.57	29381	29381	35.39	30
21	Formal education/schooling	0.93	7537	7537	77.73	60
22	Homework/assignments	0.05	544	544	84.19	90
23	Additional classes	0.02	134	134	143.97	150
24	Library/learning resources	0.48	3801	3801	47.14	30
25	Learning new skills	0.17	1393	1393	32.60	30
31	Food preparation	9.42	78595	77930	49.54	30
32	Household upkeep	2.60	21636	21636	40.88	30
33	Care for textiles/clothing	0.07	601	601	62.83	60
34	Shopping/procurement	1.62	13370	13370	50.38	60
35	Household management	0.11	826	826	64.95	60
36	NA	0.08	730	730	49.24	30
37	NA	0.50	4183	4183	63.76	60
38	NA	0.36	2972	2972	35.30	30
39	NA	0.12	1017	1017	66.44	60
41	Physical childcare	3.09	25962	25913	48.12	30
42	Teaching/supervising children	0.05	437	437	45.14	30
43	Accompanying children	0.02	136	136	56.70	30
44	Transporting children	0.10	906	906	43.45	30
49	Other childcare	0.01	105	105	61.38	60
51	Physical care adults	0.06	484	483	69.30	60
52	Healthcare support adults	0.01	95	95	113.23	120
53	Administrative help adults	0.00	20	20	173.06	150
54	Accompanying adults	0.01	118	118	57.66	30
59	Other adult care	0.00	24	24	119.70	105
61	Participating in culture	2.02	16776	16765	134.83	150
62	Attending cultural events	1.59	13276	13276	90.30	90
63	Visiting heritage sites	0.15	1289	1289	120.35	120
64	Hobbies/crafts	1.54	12925	12925	33.86	30
69	Other culture/leisure	0.02	158	158	104.93	90
71	Physical exercise/sports	7.51	61259	61154	54.73	30
72	Outdoor recreation	0.10	819	819	135.06	120
73	Hunting/fishing	0.00	33	33	68.44	60
74	Sports events attendance	1.89	16850	16850	40.27	30
75	Sports travel	0.38	3474	3473	50.55	30
79	Other sports/outdoor	0.01	129	129	107.79	120
81	Socializing in person	0.04	366	366	104.42	90
82	Socializing remotely	1.55	12621	12621	86.39	90
83	Participating in events	0.64	5451	5451	54.69	60
84	Attending ceremonies	7.13	60882	60849	60.50	60
85	Civic participation	4.48	38310	38309	54.94	30
86	Organizational meetings	0.08	653	653	41.84	30
89	Other social activities	0.04	338	338	82.06	60
91	Personal hygiene	13.27	110342	110342	222.08	180
92	Eating/drinking	17.29	144766	144766	32.71	30
93	Sleeping/napping	10.74	89778	89778	34.25	30

94	Healthcare	0.19	1557	1557	46.32	30
95	Receiving personal services	0.13	972	972	38.71	30
99	Other personal care	0.08	642	642	54.44	30

Document compiled: November 13, 2025 at 17:28

Note: All statistics use 10% random sample with adjusted weights. Percentages represent population estimates. Sample sizes (N) refer to the 10% sample and should be multiplied by 10 for full dataset estimates.